

Marketing Channel Effectiveness Study

Segmenting Customers and Profiling Acquisition Channels

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Problem Overview

Our company is preparing to launch a new product and must strategically allocate its marketing budget. However, historical spending across channels (e.g., Referral, Paid Ads, Social Media) is based on volume of acquisitions, not the value of the customers acquired. This project will use clustering analysis on historical customer data, including channel, cost of acquisition, and revenue, to identify which channels have historically attracted the most profitable and high-value customer segments. The goal is to provide a data-driven strategy for the new launch, ensuring the marketing budget is invested in channels most likely to attract valuable early adopters and maximize return on investment, rather than just maximizing initial clicks.





Motivation

Marketing departments often lack a clear understanding of which channels drive truly valuable customers.

This leads to budget being wasted on channels that generate high volume but low value, while under-investing in channels that drive high-value customers.

This project aims to solve that problem with data-driven evidence.

Objectives

New Launch

N

Market research-Target audience

Enhance CAC

Reduce the cost of acquiring high-value customers

Market ROI

Optimizing marketing spend to acquire our most profitable customers.

SCOPE OF THE STUDY

1.Objective of the Study:

The primary objective of this study is to enhance ROI,New launch Market research by creating a data-driven strategy for customer acquisition.

2. Data Key Variables:

The analysis will focus on variables directly related to customer acquisition, including revenue, conversion_rate, and channel. The study will also utilize new, engineered features such as ROI, Profit Margin, and Cost per Conversion.

3. Methodology:

Phase 1: Unsupervised Learning (Clustering): We will use K-Means clustering to perform customer segmentation. The optimal number of clusters will be determined using the Elbow and Silhouette methods.

Phase 2: Supervised Learning (Classification): We will use a Decision Tree model to classify customer channels based on the clusters identified in Phase 1. This will allow us to link customer value to acquisition channels.

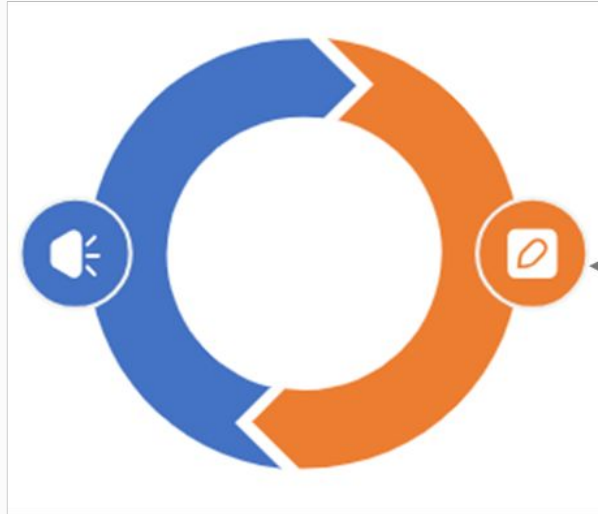
Deliverables that includes customer segment and channel recommendations, as well as a presentation summarizing the findings and strategic recommendations.

Strategy

Customer Segmentation:
Clustering(From choosing k to validation)

To identify and understand our most valuable customers.

Driver- Average _ROI



Channel Diagnosis
Classification(Tree Mapping)

To determine which channels are most effective at acquiring these valuable segments.

Driver- Channels



01

Customer Segmentations

using clustering

Customer Segmentation

Goal

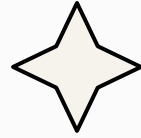
To move beyond traditional, one-size-fits-all marketing by partitioning our customer base into distinct segments based on their financial behavior.

Objective

To identify distinct customer segments based on profitability and acquisition cost.

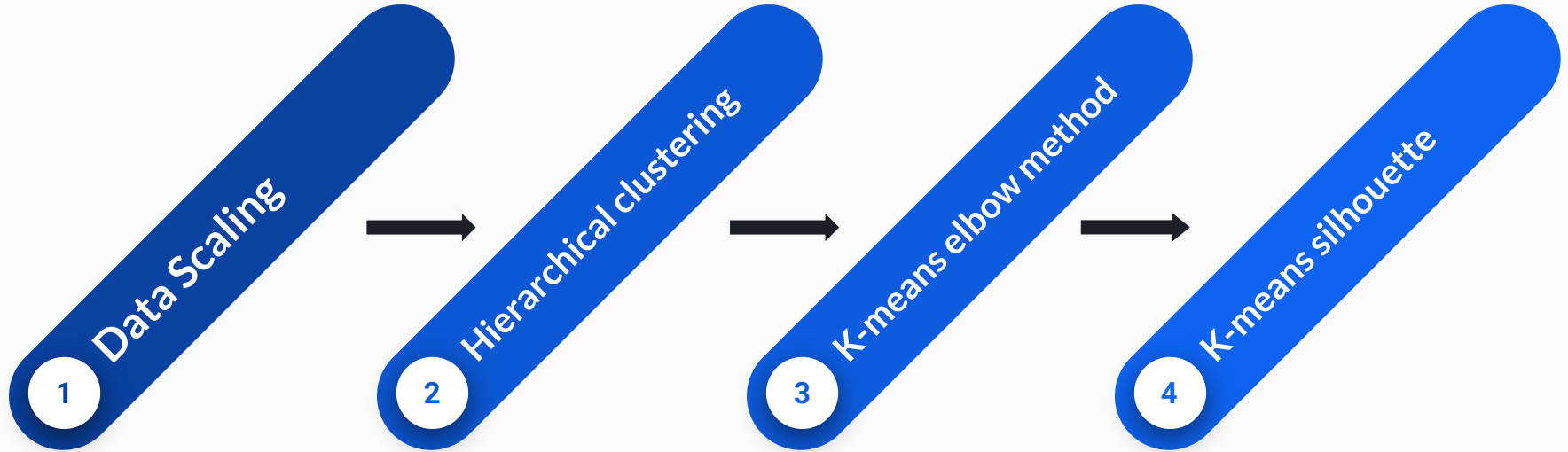
Key Metrics Used:

- Return on Investment (ROI)
- Profit Margin
- Cost per Conversion
- Expected Revenue



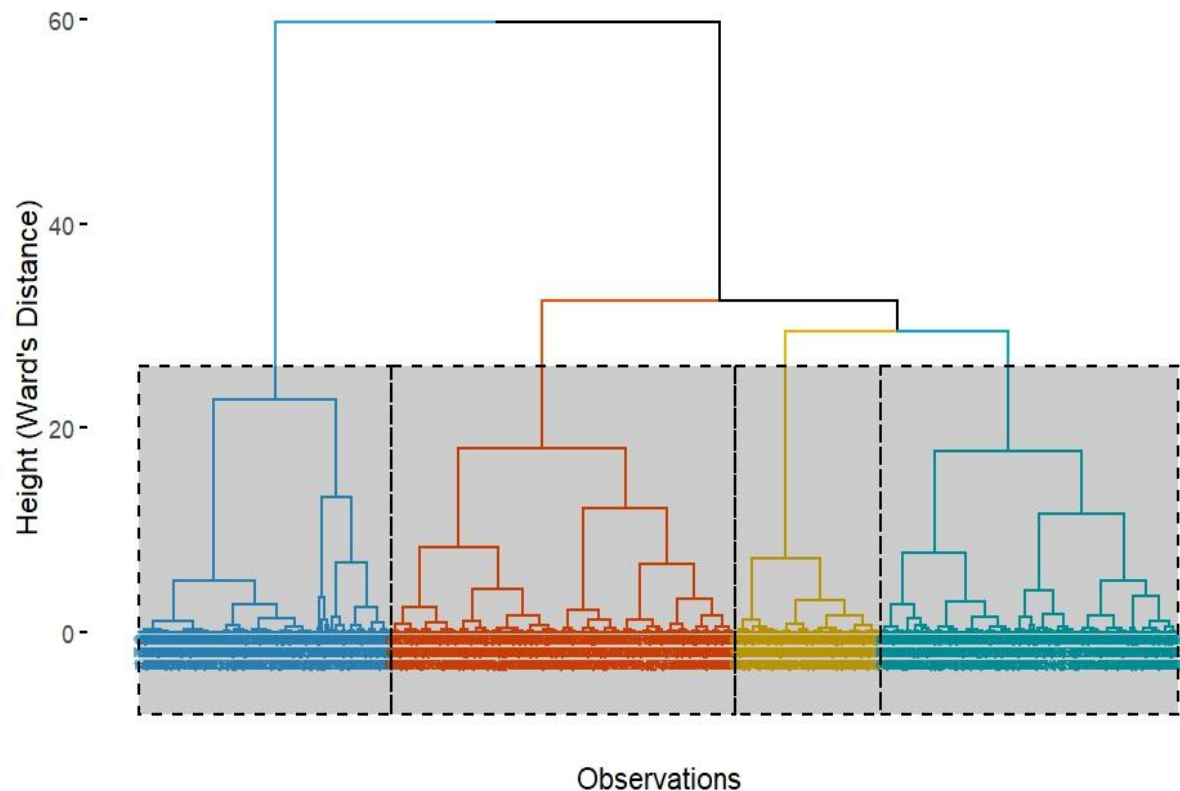
**What is the right number
of cluster size for our data
(k) ?**

Determining the optimal number of clusters



Hierarchical Clustering

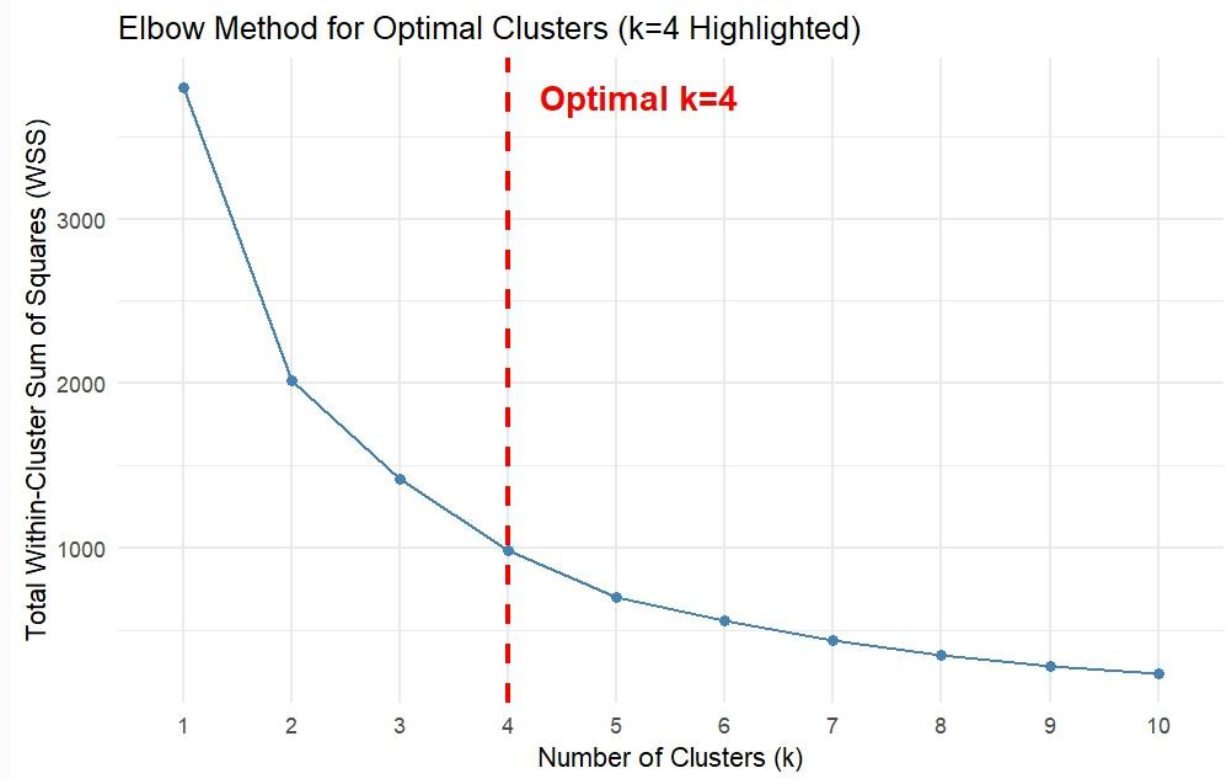
Dendrogram - Optimal k=4 Clusters Highlighted



- Reveals 4 distinct clusters
- Y-Axis (Height): Ward's Method distance. A larger height = a bigger jump in variance when clusters merge.
- Longest Vertical Lines: Indicate the largest distances between clusters, the most significant splits in the data.

K-Means Elbow Clustering

- Sharp "elbow" at $k=4$ → WSS improvement diminishes beyond this point.
- Optimal balance achieved — adding more clusters doesn't significantly improve model fit.



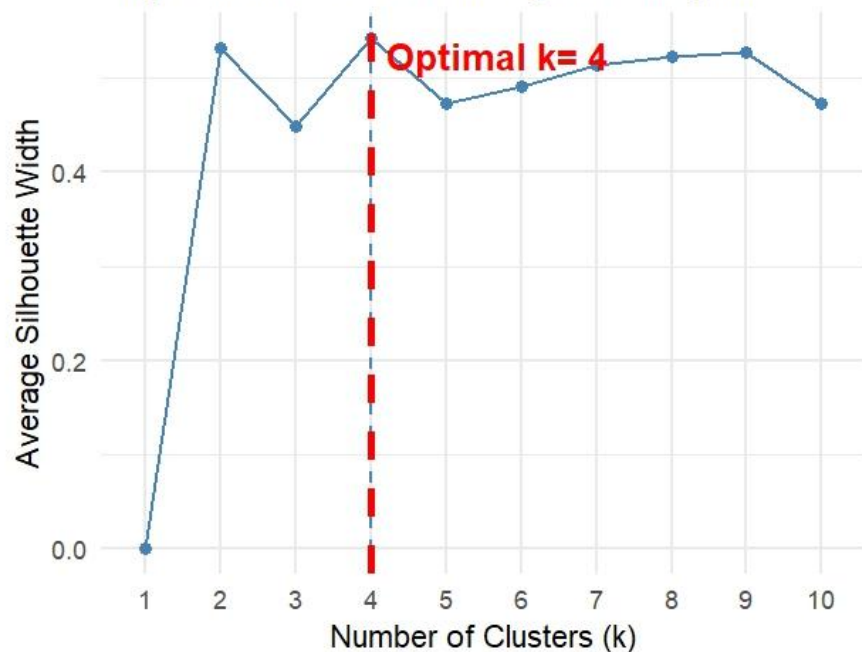
K-Means Silhouette Clustering

Impact of nstart Parameter on Optimal Cluster Selection

Higher nstart values provide more stable and reliable results

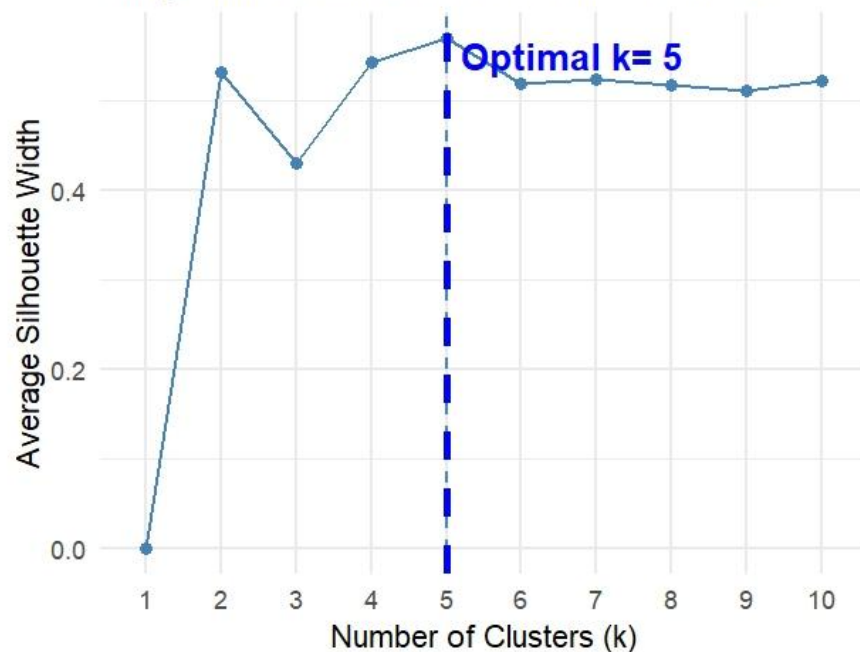
Silhouette Method (nstart = 1)

Single initialization - can converge to local optima



Silhouette Method (nstart = 25)

Multiple initializations - finds more stable solution



Comparing the nstart values

Key Insight:

- $k=4 \rightarrow 0.54$
- $k=5 \rightarrow 0.56$
- Improvement value: 0.0274 (minimal)

Thus we choose $k=4$

k	nstart = 1	nstart = 25	Difference
1	0	0	0
2	0.53	0.53	0
3	0.44	0.43	-0.018
4	0.542	0.542	0
5	0.47	0.569	0.097
6	0.49	0.51	0.02
7	0.51	0.52	0.01
8	0.52	0.51	0.004
9	0.52	0.51	0.01
10	0.47	0.52	0.05

K-Means clustering with $k=4$

Cluster	ROI	Profit Margin	COS/Conversion	Count
ROI Powerhouse	674	0.998	120	142
High ROI Optimizer	423	0.998	62.3	204
Cost-Efficient Steady Performer	205	0.994	78.6	260
High-Cost Underperformers	92.8	0.985	1863	194

Clustering Strategic Implications

01

High ROI Performers

Delivers the highest return on investment with strong margins and scalable performance.

02

High ROI Optimizer

Efficient and profitable with low acquisition cost and high conversion value.

03

Cost-Efficient Steady Performer

Reliable segment with moderate ROI and consistent cost control.

04

High-Cost Underperformers

Struggles with low ROI and high acquisition costs, requiring strategic reevaluation.



02

Channel Diagnosis

using classification

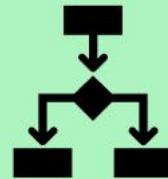
Channel Diagnosis

Goal

To build a diagnostic model that uses classification to predict acquisition channel based on customer value metrics—like cost per conversion, profit margin, and ROI, shifting our focus from volume to value in channel assessment.

Classifier

Decision Tree



Why Decision Tree ?

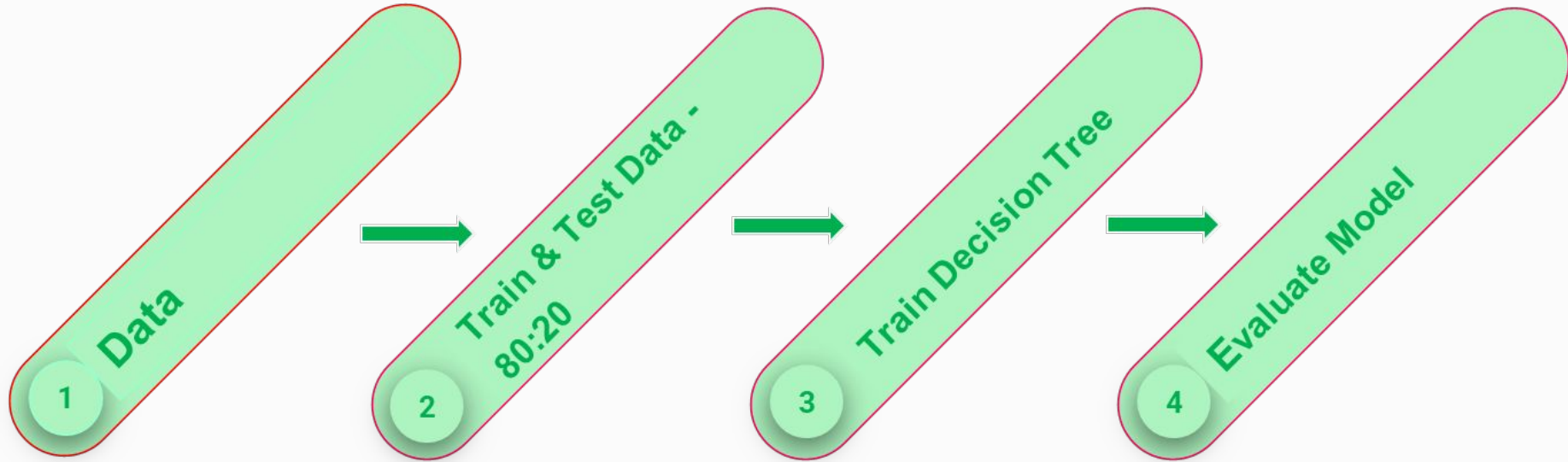
The purpose of the classification method is to predict a category or a class for a new data point. In our project, the specific purpose is to predict which marketing channel (e.g., email, social media, paid advertising) is most likely to acquire a customer based on their characteristics. This is a crucial step because it connects customer value (identified through clustering) to the most effective acquisition method.

Basics:

Predictor variables : All Features/Attributes (except target)

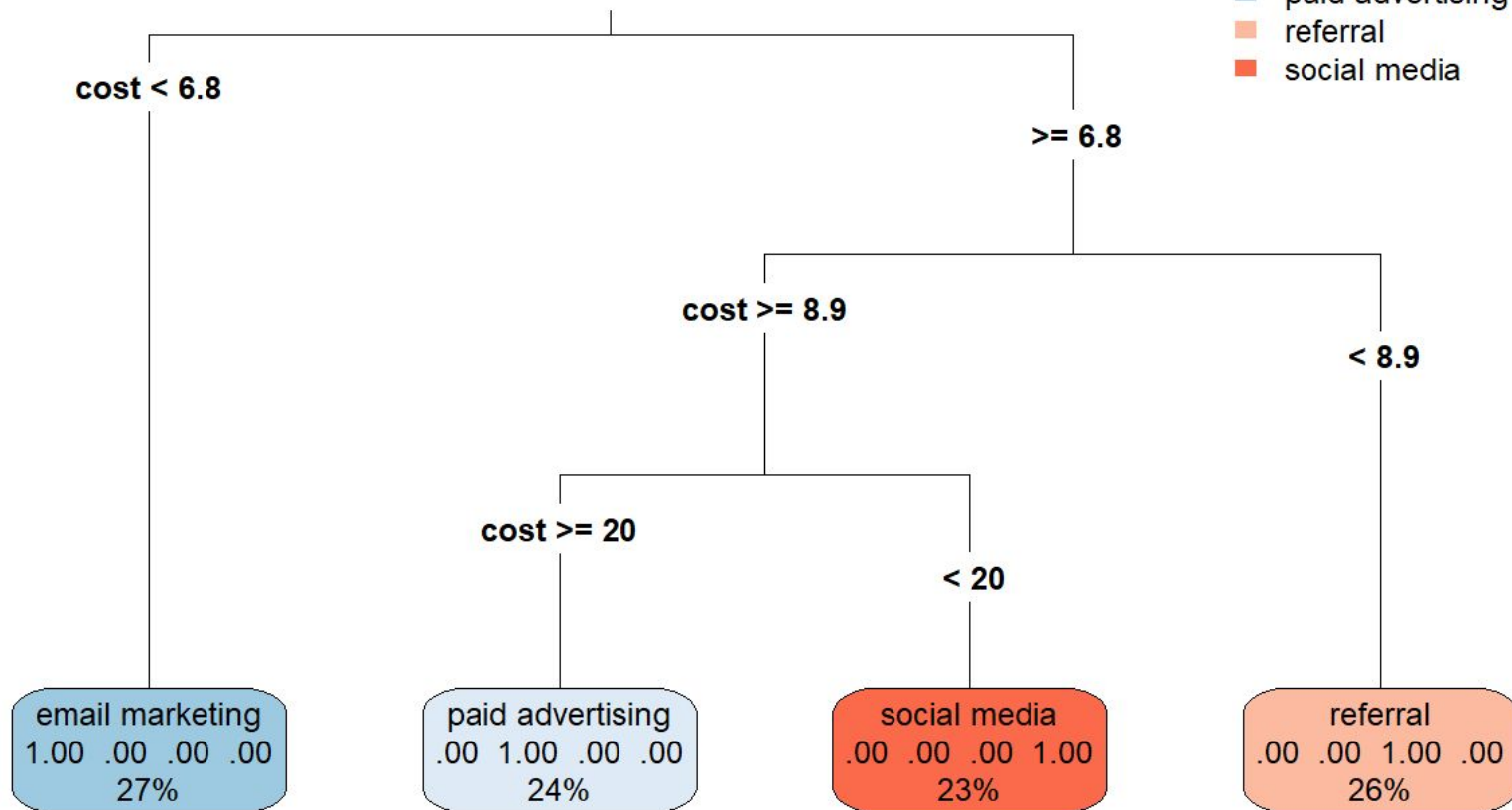
Target variable : Channel

Classification Methodology



Decision Tree: Predicting Acquisition Channel

- email marketing
- paid advertising
- referral
- social media



Classification Results

Marketing Channel	Cost per Acquisition	Key Insight
Email Marketing	< \$6.80	Confirms its status as a highly efficient channel. Indicating o healthy, low-cost marketing program.
Referral	\$6.80 - \$8.90	A cost-effective channel. This range aligns with industry expectations, as referral programs leverage existing customers to acquire new ones at a fraction of the cost of other channels.
Social Media	\$8.90 - \$20.00	Aligns with typical digital advertising costs. This CPA is moderately efficient and falls within the expected range for paid social campaigns, which can vary widely based on targeting and competition.
Paid Advertising	>= \$20.00	Highest cost channel, as expected for paid acquisition.

SUMMARY

From Phase 1
-Clustering



From Phase 2
Classification



cluster_k4_named11	count	avg_roi	avg_profit_margin	avg_cost_per_conversion	dominant_channel
1. ROI Powerhouse	142	674.2666349	0.998443661	119.7169436	email marketing
2. High ROI Optimizer	204	422.8182932	0.997532865	62.26345645	referral
3. Cost-Efficient Steady Performer	260	205.0940231	0.993907121	78.62226995	referral
4. High-Cost Underperformer	194	92.83261508	0.985053563	1863.375022	paid advertising

Recommendations

Clustering Recommendations-

- Increase marketing spend on Cluster ROI Powerhouse (ROI = 674.3, Profit Margin = 99.8%) → best-performing customers.
- Maintain balanced budgets for Cluster High-ROI Optimizer (ROI = 422.8) and Cluster 3 (ROI = 205.1) → stable but moderate performers.
- Reduce/optimize investment in Cluster High-Cost Underperformers (ROI = 92.8, Cost/Conversion = 1863.4) → poor returns.
- Impact: Higher ROI by reallocating budget towards profitable customer groups.

Classification Recommendations

- Use the decision tree model (100% accuracy) to predict acquisition channel for new customers.
- Prioritize channels with historically high ROI and profit margin.
- Automate channel allocation to reduce guesswork and improve efficiency.
- Impact: Smarter channel strategy, better budget allocation, and improved lead conversion rates

Conclusion

Clustering Analysis revealed 4 distinct customer acquisition segments:

- a. Cluster ROI Powerhouse is the most profitable (ROI 674.3, Profit Margin 99.8%, Cost/Conversion 119.7) → focus more budget here.
- b. Clusters High ROI Optimizer & Cost-Efficient Steady Performer show moderate ROI (422.8 and 205.1) with low conversion costs (62.3 and 78.6).
- c. Cluster High-Cost Underperformers performs poorly (ROI 92.8, very high Cost/Conversion 1863.4) → reduce or optimize spend.

Classification Model (Decision Tree) achieved 100% accuracy in predicting acquisition channels (Email, Paid Ads, Referral, Social Media), meaning we can reliably map customer behavior to channels and allocate marketing spend effectively.

References

1. DATA SOURCE - KAGGLE
2. TOOLS - R LANGUAGE
3. CONCEPTS - DATA MINING ALGORITHMS
4. MENTOR - Prof. Harshit Singh

THANK
YOU!